Hypothesis Testing, Separability, and Classification of Polarimetric SAR Intensity Data With Nonparametric U-Statistics

Rogério G. Negri, Alejandro C. Frery, *Life Fellow, IEEE*, and Aluísio Pinheiro

***Abstract*—Polarimetric synthetic aperture radar (PolSAR) sensors have emerged as a groundbreaking remote sensing technology. They enable the acquisition of the amplitude, phase, and orientation of electromagnetic waves across multiple polarizations. This capability provides enhanced potential for detailed environmental analysis. However, challenges such as complex data structures, non-Gaussian noise properties, and low signal-to-noise ratios pose significant barriers to the effective use of PolSAR data. Existing methods for modeling and analyzing PolSAR data are predominantly parametric and rely on assumptions that may fail under certain conditions. Aware of these limitations, this study introduces the use of U-statistics for PolSAR data analysis. Using information from the diagonal intensities of the covariance matrix, we propose a hypothesis testing mechanism to assess sample homogeneity, a top-down hierarchical separability analysis, and a U-statistics-based classification approach. The proposed procedures are validated using an ALOS-PALSAR image of the Amazonian region. The results show the robustness and effectiveness of the proposed methods, offering a reliable framework for analyzing and classifying PolSAR data under nonparametric assumptions.**

***Index Terms*—Image analysis, nonparametric statistics, radar applications, statistical analysis.**

I. Introduction

REMOTE sensing has witnessed remarkable advancements in recent years, enabling more accurate and timely environmental monitoring and analysis [1], [2]. Among these advancements, the increased availability of polarimetric synthetic aperture radar (PolSAR) sensors stands out as a significant technological breakthrough. Compared with conventional SAR sensors, PolSAR can acquire the amplitude, phase, and orientation of the electromagnetic waves reflected from targets in different transmission and reception polarizations [3], [4], [5], [6], providing enhanced capabilities for detailed environmental analysis [7].

Received 3 March 2025; revised 23 May 2025; accepted 19 August 2025. Date of publication 22 August 2025; date of current version 29 August 2025. This work was supported in part by the São Paulo Research Foundation (FAPESP) under Grant 2023/02538-0 and Grant 2024/01610-1, in part by the National Council for Scientific and Technological Development (CNPq) under Grant 305220/2022-5 and Grant 310991/2020-0, and in part by the Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPESP). *(Corresponding author: Rogério G. Negri.)*

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Digital Object Identifier 10.1109/TGRS.2025.3601821

The use of PolSAR images is challenging, among other reasons, because of the data structure (complex matrices in each pixel), their properties (the Gaussian additive noise is not valid), and the signal-to-noise ratio (typically very low) [8]. Considering these characteristics, several specific methods have been developed for PolSAR image processing [9].

Due to the inherently coherent nature of SAR imaging, statistical models have become essential tools for capturing signal variability and speckle characteristics. Over the past decades, various distributions have been extensively investigated and validated for PolSAR data analysis. Among them, the complex Wishart, K, and Fisher distributions are particularly notable. The complex Wishart distribution remains a foundational model for multilook PolSAR data, widely used in classification [10], [11], [12], change detection [13], and speckle filtering [14]. For more heterogeneous or textured scenes, the K distribution offers a robust alternative by modeling speckle and texture in a multiplicative manner [15], [16]. Subsequently, the Fisher distribution has emerged as an effective model for representing texture, particularly in segmentation and classification tasks [17], [18]. More recently, kernel-based mixture models have been proposed to flexibly capture speckle and texture components, combining the strengths of parametric and nonparametric approaches [19].

Overall, measuring the separability of observations is a crucial step in image processing and analysis of PolSAR data. Such measurements support quantitative decisions in segmentation, edge detection, and classification, among other operations. Moreover, the separability is a measure of how easy the discrimination is.

Cui et al. [20] formed superpixels aggregating statistically close observations; the authors use a mixture of Gamma distributions to describe the cluster. Nascimento et al. [21] compared the performance of edge detectors based on several stochastic distances and entropies under the complex scaled Wishart model for polarimetric data. Naranjo-Torres et al. [22] measured stochastic and geodesic distances between intensity samples under the $\mathcal{G}^0$ model. In these cases, the measures of separability have known asymptotic properties, leading to test statistics that can quantify the decision's certainty.

In contrast to the previously presented parametric perspectives, Hoeffding [23] introduced U-statistics, a fundamental concept in statistics that is widely used for estimating population parameters from samples, particularly in scenarios where assumptions about data distribution do not hold. One of the key

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properties of U-statistics is consistency, meaning that as the sample size increases, such a statistic converges in probability to the true value of the parameter. In addition, U-statistics offer significant robustness against outliers and nonnormal distributions, making them highly versatile in practical applications. Moreover, another notable advantage of U-statistics is their invariance under monotonic transformations. This property implies that applying a monotonic function to a U-statistic still results in a valid U-statistic. Such invariance is particularly useful in analyses where the relationships between variables are nonlinear.

Despite their potential and numerous appealing characteristics, studies employing U-statistics in the context of PolSAR data analysis remain absent from the current literature.

Permutation tests also identify if the samples obey the same distribution. They seek evidence that the data come from two different models. They were used by Costa et al. [24] to identify changes in the coherence of PolSAR data. U-statistics are more general because the user chooses the number of models.

With this motivation, we propose using U-statistics for hypothesis testing, separability analysis, and classification of PolSAR data based on the diagonal intensities of the observations' covariance matrix. Regarding hypothesis testing, we use a straightforward statistic that facilitates evaluating the homogeneity of a set of samples concerning the presumed groups to which they belong. This hypothesis testing subsequently forms the foundation of a top-down hierarchical separability analysis, wherein the presumed groups are iteratively segregated until the statistical homogeneity is met. We use U-statistics to classify PolSAR observations by assigning unlabeled data to the group that maximizes separability.

To evaluate the effectiveness of the proposed procedures, we perform experiments using an actual PolSAR remote sensing image acquired by the ALOS-PALSAR sensor over an Amazonian region.

In summary, the main contributions of this article are as follows.

1) The pioneering application of U-statistics concepts for PolSAR data analysis.
2) The design and formalization of novel algorithms for analyzing separability and performing classification tasks using U-statistics.
3) Comprehensive experimental validation of the proposed procedures, demonstrating their effectiveness through quantitative and qualitative assessments.
4) A freely available implementation of the proposed algorithms, fostering reproducibility and further research.

This article is structured as follows. In Section II-A, we revisit the statistical properties of the trace derived from polarimetric data, highlighting the justification for employing nonparametric statistics and introducing the formalization of data separability using U-statistics. Section III outlines the proposed procedures for analyzing data separability and performing classification using PolSAR data. Experimental results are detailed in Section IV, demonstrating the effectiveness of the proposed methods. Finally, Section V summarizes the key findings and discusses potential directions for future research.

II. Background

A. Fully PolSAR Data and Models

A multilook fully PolSAR observation in covariance matrix format has the following measurements, provided the reciprocity condition that holds:

$$\boldsymbol{C} = \begin{pmatrix} I_{\mathrm{HH}} & C_{\mathrm{HH,HV}} & C_{\mathrm{HH,VV}} \\ C^{*}_{\mathrm{HH,HV}} & I_{\mathrm{HV}} & C_{\mathrm{HV,VV}} \\ C^{*}_{\mathrm{HH,VV}} & C^{*}_{\mathrm{HV,VV}} & I_{\mathrm{VV}} \end{pmatrix} \tag{1}$$

where I_{uv} is the intensity measured when the signal is emitted in polarization u and received in polarization v, $C_{u,v}$ is the complex covariance between the channels $u \neq v$, and * denotes the complex conjugate.

Under the conditions discussed by Deng et al. [25], $\boldsymbol{C}$ follows a multilook complex Wishart distribution characterized by the following probability density function:

$$f_{\boldsymbol{C}}(z; M, \Sigma) = \frac{M^{3M}\,|z|^{M-3}\exp\left\{-M\mathrm{tr}\left(\Sigma^{-1}z\right)\right\}}{\pi^{3}\,|\Sigma|^{M}\,\Gamma(M-2)\,\Gamma(M-1)\,\Gamma(M)} \tag{2}$$

where $M > 3$ is the number of looks, $\Sigma = E(\boldsymbol{C}\boldsymbol{C}^{*T})$ is the complex covariance matrix, which is Hermitian and positive definite, tr denotes the trace of a matrix, and z spans the cone of Hermitian positive-definite matrices. We denote this situation as $\boldsymbol{C} \sim \mathcal{W}(M, \Sigma)$.

Under (2), the return from different classes differs by the covariance matrices, i.e., if $\Xi = \{\xi_1, \xi_2, \ldots, \xi_g\}$ are the classes, the return will be $\boldsymbol{C} \mid \xi \sim \mathcal{W}(L, \Sigma_\xi)$, with $\xi \in \Xi$. We assume that the number of looks L is the same for all pixels, and thus, it is known beforehand, see Anfinsen et al. [26].

The multilook complex Wishart model collapses to several interesting marginal models.

1) Gamma distributions for each element in the diagonal, i.e., $I_i \sim \Gamma(M, \mu_i)$, $1 \leq i \leq 3$, where $\Gamma(M, \mu_i)$ denotes the gamma distribution with mean $\mu_i > 0$ and shape parameter M.
2) Bivariate gamma models [27] for any two diagonal elements.
3) Trivariate gamma models [28] for the three diagonal elements.

Hagedorn et al. [28] obtained the joint probability density function of the three diagonal elements of a full-polarimetric image under the complex Wishart distribution, see [28, eq.(17)]. This function depends on an infinite sum of Gegenbauer functions (or ultraspherical polynomial functions) and the product of three Bessel functions. Such intricate expression and its associated numerical challenges preclude its use.

For this reason, we propose using a nonparametric approach that, despite not using all the capacity of the likelihood, achieves excellent results in discriminating classes.

B. U-Statistics and Data Separability

U-Statistics play a pivotal role in nonparametric estimation theory, particularly in constructing unbiased estimators with minimum uniform variance. Introduced by Hoeffding [23], U-Statistics constitute a class of statistics defined as functions

of a set of observations, making them applicable in contexts where assumptions about the data distribution do not hold or are unfeasible to use.

Consider the function $\phi\colon \mathbb{R}^{L\times k} \to \mathbb{R}$ to be applied to the kth independent and identically distributed random variables of dimension L denoted by $(X_1, X_2, \ldots, X_k)$ whose distributions are characterized by the cumulative distribution function $\overset{\circ}{F}$. We are interested in the expected value

$$\theta = \mathbb{E}\left(\phi\left(X_1, X_2, \ldots, X_k\right)\right) = \int \cdots \int \phi\left(x_1, x_2 \ldots, x_k\right) d\overset{\circ}{F}\left(x_1\right) d\overset{\circ}{F}\left(x_2\right) \cdots d\overset{\circ}{F}\left(x_k\right). \quad (3)$$

We will seek an alternative way of computing (3) because its direct evaluation is unfeasible in our application. Our approach follows Hoeffding [23].

We have access to $n > k$ random variables $(X_1, X_2, \ldots, X_n)$ of dimension L that obey the distribution characterized by $\overset{\circ}{F}$. We define an estimator for θ based on permutations of these n random variables

$$\widehat{\theta}_n = \frac{1}{\binom{n}{k}} \sum_{(i_1, i_2, \ldots, i_k)\in\mathcal{C}} \phi\left(X_{i_1}, X_{i_2}, \ldots, X_{i_k}\right) \quad (4)$$

where $\mathcal{C}$ is the set of all $\binom{n}{k}$ possible combinations of k out of the n samples.

Under mild regularity conditions, $\widehat{\theta}_n$ is an unbiased estimator for θ and, asymptotically, it follows a normal distribution [23]:

$$\sqrt{n}\left(\widehat{\theta}_n - \theta\right) \longrightarrow N\left(0, \sigma^2\right), \text{as } n \longrightarrow \infty \quad (5)$$

and $0 < \sigma^2 < \infty$.

Still, $\{\widehat{\theta}_n : n \geq k\}$ is a reverse submartingale [29], for which the following almost sure convergence holds:

$$\widehat{\theta}_n \longrightarrow \theta, \text{as } n \longrightarrow \infty. \quad (6)$$

It is worth highlighting that the presented estimator is an example of the U-statistics family. The reader will find valuable additional information on U-statistics at Pinheiro et al. [30], [31], Pinheiro [32], and References therein.

On the one hand, we have discussed U-statistics results that establish asymptotically normal and consistent estimators for populations, irrespective of their underlying distributions. They can be used, for example, to compute the mean, variance, and covariance. On the other hand, U-statistics can be used to compute measures of separability between models.

Consider two independent random samples $\boldsymbol{X} = (X_1, X_2, \ldots, X_{n_X})$ and $\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_{n_Y})$, drawn from models characterized by the cumulative distribution functions $\overset{\circ}{F}$ and $\overset{\circ}{G}$, respectively. The following is a measure of separability between such models:

$$d_\gamma\left(\overset{\circ}{F}, \overset{\circ}{G}\right) = \mathbb{E}\left(\|\boldsymbol{X} - \boldsymbol{Y}\|_\nu^\gamma\right) = \iint \|\boldsymbol{x} - \boldsymbol{y}\|_\nu^\gamma \, d\overset{\circ}{F}(x)\, d\overset{\circ}{G}(y) \quad (7)$$

where $\|\boldsymbol{x}\|_\nu$ is the $\mathscr{L}^\nu$ norm of the vector $\boldsymbol{x}$, and $\gamma > 0$. If the models coincide, i.e., if $\overset{\circ}{F} \equiv \overset{\circ}{G}$, then $d_\gamma(\overset{\circ}{F}, \overset{\circ}{G})$ measures the variability of $\overset{\circ}{F}$.

Varying γ provides flexibility in using d_γ to define measures of discrepancy between and within distributions. Small values of γ give more weight to small deviates, while large values weight to large deviates more heavily, with an effect that generalizes the well-known differences between $\mathscr{L}^\nu$ metrics, estimator robustness, and test power, among other properties.

Consider the random sample $\boldsymbol{Z} = (Z_1, Z_2, \ldots, Z_{n_Z})$. We can measure the variability of $\boldsymbol{Z}$ by computing

$$\widehat{d}_\gamma(\boldsymbol{Z}) = \frac{2}{n_{\boldsymbol{Z}}\left(n_{\boldsymbol{Z}} - 1\right)} \sum_{i=1}^{n_{\boldsymbol{Z}}-1} \sum_{j=i+1}^{n_{\boldsymbol{Z}}} \left\|Z_i - Z_j\right\|_1^\gamma. \quad (8)$$

Now, if we have two random samples $\boldsymbol{X} = (X_1, X_2, \ldots, X_{n_X})$ and $\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_{n_Y})$, we can estimate the discrepancy between their distributions with

$$\widehat{d}_\gamma(\boldsymbol{X}, \boldsymbol{Y}) = \frac{1}{n_{\boldsymbol{X}} n_{\boldsymbol{Y}}} \sum_{i=1}^{n_X} \sum_{j=1}^{n_Y} \left\|X_i - Y_j\right\|_1^\gamma. \quad (9)$$

The statistics defined by (8) and (9) are consistent estimators, so both normality (5) and consistency (6) hold, as long as $\mathbb{E}(\|\boldsymbol{X} - \boldsymbol{Y}\|_1^{2\gamma}) < \infty$.

Let us consider the need to analyze the statistical difference between the models $\overset{\circ}{F}$ and $\overset{\circ}{G}$, e.g., for classification, clustering, or edge detection. One may tackle the problem by drawing samples $\boldsymbol{X} = (X_1, X_2, \ldots, X_{n_X})$ from $\overset{\circ}{F}$, and $\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_{n_Y})$ from $\overset{\circ}{G}$, and testing the hypothesis

$$H_0: \ \overset{\circ}{F} \equiv \overset{\circ}{G} \text{ vs. } H_1: \ \overset{\circ}{F} \neq \overset{\circ}{G}.$$

With basis on the previous formalization, and adopting $d_\gamma(\cdot,\cdot)$ as U-statistic-based estimators θ as in (3) and (7), we obtain the following three population quantities:

$$\theta\left(\overset{\circ}{F}\right) = d_\gamma\left(\overset{\circ}{F}\right) \text{— the variability within } \overset{\circ}{F}$$
$$\theta\left(\overset{\circ}{G}\right) = d_\gamma\left(\overset{\circ}{G}\right) \text{— variability within } \overset{\circ}{G}$$
$$\theta\left(\overset{\circ}{F}, \overset{\circ}{G}\right) = d_\gamma\left(\overset{\circ}{F}, \overset{\circ}{G}\right) \text{— variability between } \overset{\circ}{F} \text{ and } \overset{\circ}{G}.$$

Estimator (8) applied to $\boldsymbol{X}$ ($\boldsymbol{Y}$, respectively) measures the within-sample statistics for the sample $\boldsymbol{X}$ ($\boldsymbol{Y}$, respectively) Estimator (9) is the between-sample (or cross sample) statistic. These estimators are generalized U-statistics, and under mild regularity conditions, they satisfy the properties of asymptotic normality (5), i.e., they follow an asymptotic trivariate normal distribution, and almost sure convergence (6).

One may prove that $2\theta(\overset{\circ}{F}, \overset{\circ}{G}) \geq \theta(\overset{\circ}{F}) + \theta(\overset{\circ}{G})$, with equality if and only if $\overset{\circ}{F} \equiv \overset{\circ}{G}$ (i.e., when H_0 is true) [31]. Consequently, hypothesis testing can be conducted using the following test statistic T to assess whether there is statistical evidence for $H_0\colon \overset{\circ}{F} \equiv \overset{\circ}{G}$ versus $H_1\colon \overset{\circ}{F} \neq \overset{\circ}{G}$:

$$T(\boldsymbol{X}, \boldsymbol{Y}) = 2\widehat{\theta}(\boldsymbol{X}, \boldsymbol{Y}) - \widehat{\theta}(\boldsymbol{X}) - \widehat{\theta}(\boldsymbol{Y}). \quad (10)$$

This property is the basis of our proposal.

We may proceed with this test, applying its asymptotic reasoning, which results in a two-sided normal test. Pinheiro et al. [30], [31] argued that a one-sided test is also possible in this scenario and that such a test statistic is asymptotically normal as well. There are two advantages to employing the one-sided test instead of the two-sided approach. First, there is a duality between the hypothesis

H_0: $\overset{\circ}{F} \equiv \overset{\circ}{G}$ versus H_1: $\overset{\circ}{F} \neq \overset{\circ}{G}$ and the one-sided problem $\mathbb{E}(T) = 0$ versus $\mathbb{E}(T) > 0$, where T is the aforementioned test statistic.

There are clear connections to classic distribution tests such as chi-squared tests and other Cramér–von Mises divergence statistics. The communality is that, under H_0, the rate of convergence is $q = n_X + n_Y$ whilst, under H_1, it is $(q)^{1/2}$. That is due to a natural degeneration of the first-order term under H_0, because of the canceling effects derived from the true homogeneity hypothesis. The difference is that, for these classic tests, the null distributions are not normal. Quasi U-statistics, through the careful choice of weights, which form a contrast based on an $\mathscr{L}_2$ norm, have asymptotic normal distributions under the null and alternative hypotheses.

Confronted with two samples, $\boldsymbol{X}$ of size n_X and $\boldsymbol{Y}$ of size n_Y, we want to check if there is enough evidence to keep them split, or if they are homogeneous and should be merged into a single sample $\boldsymbol{Z} = (\boldsymbol{X}, \boldsymbol{Y})$ of size $q = n_X + n_Y$. We apply the test statistic

$$T(\boldsymbol{Z}) = \sum_{i,j=1}^{q} \eta_{ij} \left\| Z_i - Z_j \right\|_1^{\gamma} \tag{11}$$

with weights

$$\eta_{ij} = \begin{cases} 1, & \text{if } Z_i \text{ and } Z_j \text{ are in different groups} \\ -\dfrac{(q - n_\star)}{(n_\star - 1)}, & \text{if } Z_i \text{ and } Z_j \text{ are in the same group} \end{cases} \tag{12}$$

with $n_\star$ is the cardinality of the group that contains Z_i and Z_j, i.e., either n_X or n_Y. These weights create a contrast on the within- and between-groups separability measures with the good statistical properties stated above and detailed in [30] and [31].

Regarding the test statistic defined in (11), although U-statistics theory provides asymptotic formulations for hypothesis testing and parameter estimation, the bootstrap offers a simpler and more practical alternative, which remains effective even for moderate sample sizes [30].

III. U-Statistics for Homogeneity Test and Image Classification

As discussed in Section I, there are a few powerful statistical procedures for analyzing polarimetric observations without relying on distributional assumptions, which can become unrealistic or even constrained by demanding theoretical prerequisites. For this reason, we propose the nonparametric U-statistics, introduced in Section II-B, for analyzing the information carried by the three intensity channels in polarimetric data. Section III-A presents a procedure for assessing the homogeneity of groups, which is then extended into another procedure for hierarchically evaluating the data separability. Subsequently, a novel classification procedure based on U-statistics is formalized in Section III-B.

A. Homogeneity Test

The discussions presented in Section II-B provide a formal foundation for comparing samples or their models. We discussed test statistics, which are asymptotically normal under both the null and alternative hypotheses. These theoretical properties enable the implementation of hierarchical clustering procedures based on tests with known significance, among other advantages. More specifically, the test results can be interpreted as follows. If H_0 is rejected, we conclude that there is statistical evidence (at the α significance level) that there are at least two different groups. Conversely, if H_0 is not rejected, we conclude that no statistical evidence (at the α significance level) exists that the groups are not homogeneous.

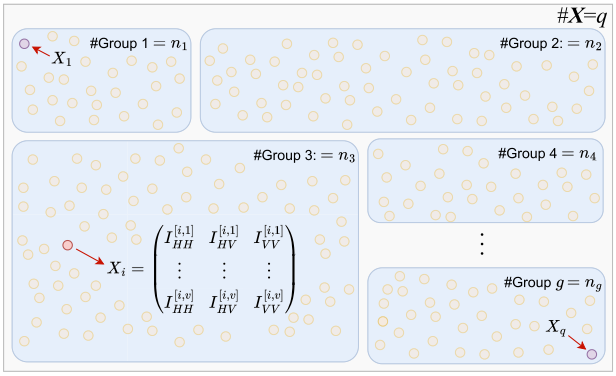


Fig. 1. Overview of the sample set $\boldsymbol{X}$.

To analyze the heterogeneity (or homogeneity) of a $\boldsymbol{X} = (X_1, X_2, \ldots, X_q)$, consider that this set contains observations drawn from g presumed groups with cardinalities $n_1, n_2, \ldots, n_g$, such that $\sum_{i=1}^{g} n_i = q$. These groups are determined based on prior information, e.g., with training samples, rather than being statistically established through formal tests. The observations, denoted by X_i, consist of v measurements corresponding to the diagonal elements $(I_{\mathrm{HH}}, I_{\mathrm{HV}}, I_{\mathrm{VV}})$ of the covariance matrix (1). For sake of conciseness, we may denote the jth measurement of X_i by $(I_{\mathrm{HH}}^{[i,j]}, I_{\mathrm{HV}}^{[i,j]}, I_{\mathrm{VV}}^{[i,j]})$. Under this assumption, we have $X_i \in \mathbb{R}^{x \times 3}$ conveniently represented by the entity

$$X_i = \begin{pmatrix} I_{\mathrm{HH}}^{[i,1]} & I_{\mathrm{HV}}^{[i,1]} & I_{\mathrm{VV}}^{[i,1]} \\ \vdots & \vdots & \vdots \\ I_{\mathrm{HH}}^{[i,v]} & I_{\mathrm{HV}}^{[i,v]} & I_{\mathrm{VV}}^{[i,v]} \end{pmatrix}.$$

Fig. 1 summarizes the dimensionality and organization of the sample set $\boldsymbol{X}$.

Based on the input sample set $\boldsymbol{X}$ and a suitable choice for the parameter γ, the statistic T can be straightforwardly computed using (11). Since the distribution of this statistic is unknown, a Monte Carlo simulation with B bootstrap replications is used to estimate the required distribution and assess the occurrence of T. Finally, using the distribution obtained from the Monte Carlo simulation, the p-value is computed to evaluate the likelihood of observing a value equal to or more extreme than the observed T. Based on a predefined significance level α, a conclusion about the separability of $\boldsymbol{X}$ is drawn. The described test procedure is formalized in Algorithm 1.

Assume a scenario with two groups, i.e., $g = 2$. Interpreting the result of the discussed test is straightforward. If H_0 is not rejected, the initial two groups are merged into a single group, as no statistical evidence supports their separability.

Algorithm 1 U-Statistics-Based Separability Analysis

Input: $\boldsymbol{X} = (X_1, \ldots, X_q)$ containing g presumed groups
Input: a value for parameter γ and the significance level α
Input: the number B of bootstrap replications
Data: $T = \sum_{1 \le i,j \le q} \eta_{ij} \| X_i - X_j \|_1^\gamma$
for $1 \le b \le B$ **do**
  draw a bootstrap sample $\boldsymbol{X}^{(b)} = (X_1^{(b)}, X_2^{(b)}, \ldots, X_q^{(b)})$
  $T^{(b)} = \sum_{1 \le i,j \le q} \eta_{ij} \| X_i^{(b)} - X_j^{(b)} \|_1^\gamma$
Output: p-value of T based on the observed bootstrap test statistics $(T^{(1)}, T^{(2)}, \ldots, T^{(B)})$

Conversely, if H_0 is rejected, the two groups remain distinct, as they are statistically separable.

In a more general scenario where $g \ge 3$, if H_0 is not rejected, it is statistically established that there is no separability between any pairs of groupings. Consequently, all the groups are merged into a single group. Conversely, if H_0 is rejected, it indicates that at least two groups are statistically separable. However, it is important to note that rejecting H_0 does not provide information on the exact number of distinct groups present.

For situations where $g \ge 3$ and H_0 is rejected, the analysis outlined in Algorithm 1 can be extended into a recursive top-down hierarchical scheme optimized using the test statistic T and the within-groups variability

$$W = \sum_{\substack{1 \le i,j \le n_g \\ i \ne j}} \eta_{ij} \left\| X_i - X_j \right\|_1^\gamma .$$

Initially, for a given sample set $\boldsymbol{X}$, which contains g presumed groups and a fixed parameter configuration (γ, B, α), a homogeneity analysis is conducted using Algorithm 1. If H_0 is not rejected, the process concludes that all presumed groupings in $\boldsymbol{X}$ can be merged into a single group. However, if H_0 is rejected, $\boldsymbol{X}$ is bipartitioned according to all possible combinations of its presumed g groups. Each resulting "two-group interpretation" is then evaluated using Algorithm 1, and the respective p-value is obtained. For all bipartitions that produce separable groups (i.e., where H_0 is rejected), the corresponding T, W, and W/T values are computed. The bipartition that minimizes W/T is selected and denoted by $(\boldsymbol{X}_L)$–$(\boldsymbol{X}_R)$, alluding to the resulting "left" and "right" portions.

The above-described process is then recursively applied to $\boldsymbol{X}_L$ and $\boldsymbol{X}_R$, thereby generating a hierarchical structure based on the separability of the samples/groups. The generalized scheme is formalized in Algorithm 2.

When testing the proposed groups using quasi U-statistics, there are two primary approaches for assessing the null hypothesis: employing the theoretical distribution or using bootstrap resampling. In the first approach, when there is clear evidence that the asymptotic distribution closely approximates the theoretical distribution, the asymptotic Gaussian distribution can be used. In this case, hypothesis testing (e.g., rejecting H_0 at a significance level α or computing the corresponding

Algorithm 2 Hierarchical Separability Analysis Based on U-Statistics

Input: $\boldsymbol{X} = (X_1, \ldots, X_q)$, containing g presumed groups.
Input: a value for parameter γ and the significance level α
Data: compute the p-value by applying Algorithm 1 to $\boldsymbol{X}$
if *p-value* $< \alpha$ **then**
  compute all *bi-partitions* on $\boldsymbol{X}$ based on its g presumed groups
  foreach *bi-partition of* X **do**
    $T = \sum_{1 \le i,j \le n_g} \eta_{ij} \| X_i - X_j \|_1^\gamma$
    $W = \sum_{\substack{1 \le i,j \le n_g \\ i \ne j}} \eta_{ij} \| X_i - X_j \|_1^\gamma$
  select the bi-partition $(\boldsymbol{X}_L)$-$(\boldsymbol{X}_R)$ with smallest W/T
  apply Algorithm 2 using $\boldsymbol{X}_L$ as input
  apply Algorithm 2 using $\boldsymbol{X}_R$ as input
else
  Output: show the presumed groups in $\boldsymbol{X}$.

p-value) is performed using the standard normal distribution or even tabular values. Conversely, in more exploratory scenarios where the adequacy of the Gaussian approximation is less certain, bootstrap resampling may be employed. Decisions can then be made either by fixing a significance level α or by calculating empirical p-values. In either case, the strong asymptotic properties of U-statistics support the validity of both approaches.

To compute the bootstrap p-value associated with the test statistic T, we generate B replicate samples $\boldsymbol{X}^{(b)}$ $(b = 1, \ldots, B)$ by resampling with replacement from the original data, maintaining group sizes. For each replicate, the test statistic $T^{(b)}$ is computed. The p-value is then estimated as the proportion of $T^{(b)}$ values greater than or equal to the observed T.

B. U-Statistics for Image Classification

Land use and land cover (LULC) classification is among the most common environmental applications, typically combining data classification algorithms with remotely sensed data. For LULC mapping using SAR sensors, their independence from sunlight and favorable atmospheric conditions makes them particularly advantageous. However, SAR data pose significant challenges due to their low signal-to-noise ratio and the presence of speckle that results from the coherent imaging process. In light of these issues, the development of new classification models for SAR data remains a persistent and active area of research.

Typical supervised image classification approaches in the literature include the use of the maximum likelihood classifier (MLC) [33], Mahalanobis distance classifier (MDA) [34], support vector machines (SVMs) [35], and multilayer perceptron (MLP) [36] in a pixel-based framework [37]. Alternatively, region-based approaches can be employed, where pixels are initially aggregated into homogeneous objects using segmentation techniques and subsequently classified individually [38].

A simple and effective classification rule in such approaches involves using a statistical distance between the representative distribution of the LULC classes of interest and an unlabeled region [39].

Formally, let $\mathcal{I}$ be an image defined on the grid $\mathcal{S} \subset \mathbb{N}^2$, where the pixel values are elements of the attribute space $\mathcal{X} = \mathbb{R}^L$. We denote $\mathcal{I}(s) = \boldsymbol{x}_s$ to indicate that a pixel $s = (s_1, s_2) \in \mathcal{S}$ of $\mathcal{I}$ has the attribute $\boldsymbol{x}_s$.

A pixel-based classification is a function $C\colon \mathbb{R}^L \to \mathcal{Y}$, where $C(\boldsymbol{x}_s) = y_s$ indicates that pixel $s \in \mathcal{S}$ is assigned to a class label $y_s \in \mathcal{Y} = \{1, 2, \ldots, g\}$. The classifier C, in a supervised learning paradigm, is trained using examples from a training set $\mathcal{D} = \{(\boldsymbol{x}_i, y_i) \in \mathbb{R}^L \times \mathcal{Y} : i = 1, \ldots, q\}$. As a result of applying C to the image $\mathcal{I}$, a map $\mathcal{M}$ is obtained, representing the occurrence of the LULC classes observed in $\mathcal{D}$ across the image support $\mathcal{S}$.

Similarly, a region-based classification process involves assigning a class label y to all pixels within a spatially continuous region $\mathcal{R}_i \subset \mathcal{S}$, which is derived through a segmentation procedure. In this context, the fundamental distinction between pixel-based and region-based approaches lies in the unit of analysis, i.e., pixel or region. Consequently, the classification methods commonly found in the literature can be adapted for use in either scheme.

In the context of the above discussion, the use of U-statistics emerges as a promising alternative for SAR data classification, mainly when the statistical properties of the attributes are unknown or hard to embed into the building of C, e.g., for the diagonal elements of polarimetric observations. The independence from distributional assumptions and the high sensitivity to data separability are key factors motivating the development of a classification approach for SAR data based on this type of statistic.

According to the discussion in Section III-A, the statistic T quantifies the "homogeneity/separability level" of a given set of samples composed of (or presumed to be) different groups. Consequently, for a training set containing observations from distinct groups (or classes), an observation is assigned to a group when its inclusion maximizes the separability among all groups.

For classification purposes in a pixel-based fashion, each position s is analyzed using the distributional information contained in its neighboring observations X_s. For instance, define the neighboring observations within a square of side 2ρ

$$X_s = \left\{\boldsymbol{x}_j \colon \max\left\{|s_1 - j_1|, |s_2 - j_2|\right\} \leq \rho; \forall\, j = (j_1, j_2) \in \mathcal{S}\right\}.$$

Accordingly, given a training set $\mathcal{D} = \{(\boldsymbol{x}_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \ldots, q\}$, with observations from g distinct classes/groups, and a fixed ρ, such a training set becomes equivalent to a sample set $\boldsymbol{X} = (X_1, X_2, \ldots, X_q)$, with group membership specified by y_i.

In order to assign a class/group to X_s, let us define $\overline{\boldsymbol{X}}_{s,\ell} = (X_s, \boldsymbol{X})$ as a set of samples; such elements' group pertinence is the same for those in $\boldsymbol{X}$ and X_s being assigned to the ℓth group (i.e., $y_s = \ell$). By considering $\overline{\boldsymbol{X}}_{s,\ell}$, one may compute

$$T\left(\overline{\boldsymbol{X}}_{s,\ell}\right) = \sum_{X_i, X_j \in \overline{\boldsymbol{X}}_{s,\ell}} \eta_{ij} \left\| X_i - X_j \right\|_1^{\gamma}. \tag{13}$$

Finally, with basis on these entities, we define the classifier $C\colon \overline{\mathcal{X}} \to \mathcal{Y}$ expressed by

$$C\left(\overline{\boldsymbol{X}}_{s,\ell}\right) = \arg\max_{\ell=1,\ldots,c} \{T\left(\overline{\boldsymbol{X}}_{s,\ell}\right)\} \tag{14}$$

where $\overline{\mathcal{X}}$ is the domain of sets of samples.

Algorithm 3 U-Statistics-Based Image Classification

Input: Image $\mathcal{I}$ with support S and pixels assigned to the intensities computed from a polarimetric observation.
Input: $\mathcal{D} = \{(\boldsymbol{x}_i, y_i) \in \mathbb{R}^L \times \mathcal{Y} \colon i = 1, \ldots, q\}$
Input: the value for parameters γ and ρ.
Data: determine $\boldsymbol{X} = (X_1, X_2, \ldots, X_q)$ using $\mathcal{D}$
foreach $s \in \mathcal{S}$ **do**
  for $1 \leq \ell \leq c$ **do**
    $\overline{\boldsymbol{X}}_{s,\ell} = (X_s, \boldsymbol{X})$
    $T(\boldsymbol{X}_{s,\ell}) = \sum_{X_i \in \overline{\boldsymbol{X}}_{s,\ell}} \sum_{X_j \in \overline{\boldsymbol{X}}_{s,\ell}} \eta_{ij} \|X_i - X_j\|_1^{\gamma}$
  $\mathcal{M}(s) = C(T(s, \ell))$
Result: $\mathcal{M}$

According to C, the position s will be assigned to ω_ℓ that maximizes the separability in $\overline{\boldsymbol{X}}_{s,\ell}$. Finally, the resulting classification output is the map $\mathcal{M}$ such that $\mathcal{M}(s) = C(\overline{\boldsymbol{X}}_{s,\ell})$. Algorithm 3 shows the proposed U-statistic-based approach for image classification.

IV. Experiments

A. Experimental Design

To evaluate the effectiveness of the U-statistics-based procedures proposed in Section III, experiments were conducted using an actual remote sensing image acquired by the ALOS-PALSAR over an Amazonian region. The area is rich and complex, prone to dense cloud cover, and we have access to several LULC samples. Further details regarding the data and experimental setup are provided in Section IV-B.

Specifically, with basis on the prior knowledge of the LULC samples, an analysis of separability was performed (see Algorithm 2) under two distinct cases: initially (Case I) set of seven distinct and separable groups, defined by means of the LULC classes/types, for which the null hypothesis H_0 is expected to be rejected in any attempt to merge two groups; posterior (Case II) set of three groups, where two of them are nonseparable, implying the nonrejection of H_0. Throughout the experiment, the statistical quantities that permeates the formalization presented in Section III-A are discussed.

Subsequently, the results from this first experiment will provide a foundation for ensuring a set of separable groups from distinct LULC classes. These groups will be utilized to classify the entire scene using the classification procedure proposed in Section III-B. The classification results are then compared with established methods from the literature and evaluated in terms of overall accuracy (OA) and kappa coefficient [40], computed using reference samples. We used for comparison MLC and MDA methods assuming that the classes can be described by multivariate Gaussian models, SVM equipped with radial basis kernel function, MLP, and

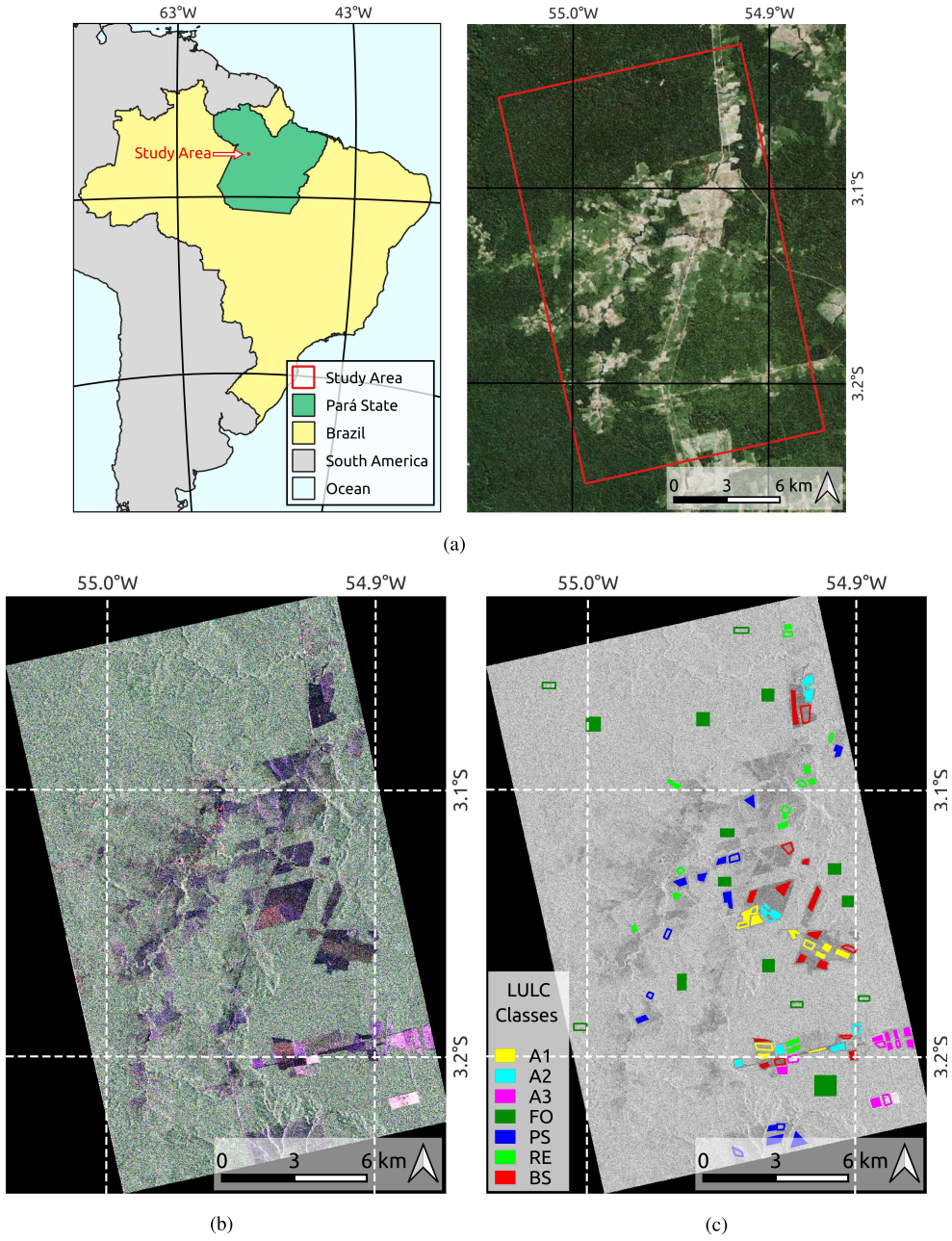


Fig. 2. Study area, the intensities of a PolSAR image, and the spatial distribution of the LULC samples used in the study. (a) Study area location. (b) ALOS-PALSAR image in RGB color composition (HH, HV, and VV). (c) LULC samples.

MSDC based on the Bathacharrya distance. The parameters required by SVM (e.g., penalty term and kernel flexibility) and MLP (e.g., number of layers, activation function, learning rate, and momentum constant) are optimized using a Grid-Search procedure [41].

We present a separability analysis (see Section IV-C) before the classification (Section IV-D). Both include an ablation study of the parameters γ [the exponent of the U-statistic (13)] and ρ (that controls the neighborhood size), evaluating their impact on data separability. The tested values for γ are $\{0.1, 0.2, 1.9\}$, while ρ ranges from $\{1, 2, \ldots, 8\}$, corresponding to regular neighborhoods of 3×3, 5×5, and 17×17 pixels. Regarding the number of bootstrap replications (B) and the significance level (α) required in Algorithms 1 and 2, these were set to 2000 and 5%, respectively.

The experiments were implemented using interactive data language (IDL) version 9.0. The code is freely available at: https://github.com/rogerionegri/ustat

B. Dataset

We used a PolSAR image acquired by the ALOS-PALSAR sensor on March 13, 2009. This image contains the HH, HV, and VV intensity polarizations [i.e., the diagonal of (1)], L band, with an approximate resolution of 20×20 m after a 3×3 spatial multilook process. The scene corresponds to a

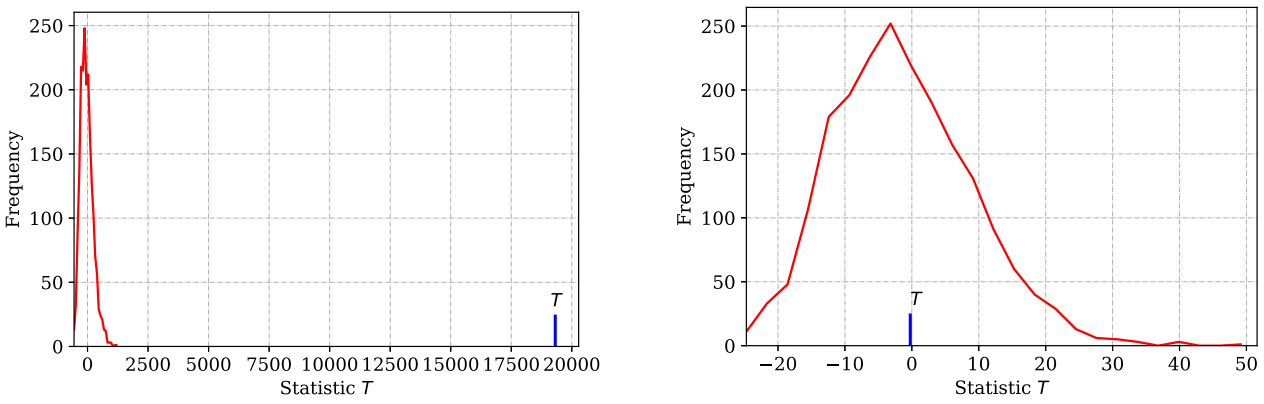


Fig. 3. Computed statistic T, with $\rho = 5$ and $\gamma = 1.0$, in comparison to the distribution computed from the Monte Carlo simulation. (a) Case I. (b) Case II.

TABLE I
SUMMARY OF THE LULC CLASSES SAMPLES

LULC Classes	Training		Testing	
	Polygons	Pixels	Polygons	Pixels
Agriculture 1 (A1)	4	3669	8	7455
Agriculture 2 (A2)	4	2902	8	6731
Agriculture 3 (A3)	3	2332	8	7049
Primary Forest (PF)	3	5430	10	29306
Pasture (PS)	5	3334	10	12866
Regeneration (RE)	5	2570	10	7307
Bare Soil (BS)	5	5384	11	13352

region near the Tapajós National Forest, State of Pará, Brazil [see Fig. 2(a)].

A fieldwork campaign conducted in September 2009 identified the following LULC types: primary forest (PF), regeneration (RE), pasture (PS), and bare soil (BS), and three Agricultural areas in distinct growth stages (A1–A3) of different crop types or growth stages. Table I summarizes the LULC samples, while Fig. 2(b) and (c) presents a color composition of the ALOS-PALSAR image and the corresponding LULC samples, respectively. With respect to these samples, solid polygons represent the training samples, and empty polygons denote the test samples used in the classification experiment (see Section IV-D). However, this distinction between training and testing samples is not considered in the separability analysis experiment presented in Section IV-C.

C. Separability Analysis

According to the experimental design outlined in Section IV-A, the proposed separability analysis procedure is applied to two distinct scenarios. The first case considers all seven available LULC samples/classes: PF, RE, PS, BS, and three agricultural classes (A1–A3). In this scenario, separability among the classes is expected. In contrast, the second case assumes a subset of classes limited to A2, PFa, and PFb. Here, PFa and PFb are derived from a random split of class PF, and their separability is not expected.

Regarding the separability of Case I, regardless of the parameter configuration set for γ and ρ, the returned p-values

TABLE II
T AND CRITICAL VALUES OBSERVED FOR CASE I WITH $\rho = 5$

γ	T	Upper quantiles						
		0.8	0.9	0.95	0.975	0.99	0.999	0.9999
0.1	1763	47.7	76.2	99.0	117.5	139.3	202.3	214.2
0.2	3453	54.3	84.4	118.3	140.9	173.0	222.4	243.3
0.3	5121	64.2	107.9	142.7	173.9	206.2	271.1	324.1
0.4	6806	76.1	129.9	178.1	222.5	274.0	396.4	466.7
0.5	8546	94.4	172.5	232.4	279.8	348.8	552.5	615.9
0.6	10370	118.3	190.3	270.4	326.6	446.0	654.2	722.2
0.7	12309	134.6	227.4	307.8	387.2	485.7	651.5	798.4
0.8	14395	163.9	276.0	379.2	464.3	563.6	809.8	920.9
0.9	16657	181.1	307.4	448.1	556.0	695.6	1142	1304
1.0	19128	207.3	353.2	499.7	600.9	780.6	1150	1382
1.1	21845	232.2	415.3	563.3	719.5	948.4	1483	1729
1.2	24846	238.8	438.8	650.8	891.6	1144	1728	1924
1.3	28175	287.5	520.0	685.7	852.9	1154	1560	2038
1.4	31880	350.2	631.4	844.3	1025	1258	1901.2	2287
1.5	36016	395.8	713.6	975.1	1196	1445	2236	2874
1.6	40644	466.0	801.6	1116	1387	1781	2305.4	2896
1.7	45833	487.3	887.8	1236	1508	1997	2845	3083
1.8	51664	546.3	1017	1432	1840	2475	3874	6130

from the homogeneity test (see Algorithm 1) are all equal to 0.005 (the smallest possible value for a bootstrap resample of $B = 2000$). This indicates that, at a 5% significance level, the seven-class LULC sample set is heterogeneous and contains at least one pair of classes originating from distinct groups. The validity of this result is corroborated by Table II (presented for $\rho = 5$ to ensure compactness while using an intermediate parameter value), which highlights the significant difference between the statistic T and the upper quantiles of the distribution obtained from the Monte Carlo simulation. This comparison is further illustrated in Fig. 3(a), assuming $\rho = 5$ and $\gamma = 1.0$.

The hierarchical scheme described in Algorithm 2 was applied for an in-depth separability analysis. For the sake of compactness, the parameters were fixed at $\rho = 5$ and $\gamma = 1.0$. Table III presents the most prominent bipartitions, sorted by the statistics T, W, W/T, and p-values, across different hierarchical levels. Initially, it is observed that the classes/groups A3, PF, and RE exhibit higher similarity when compared

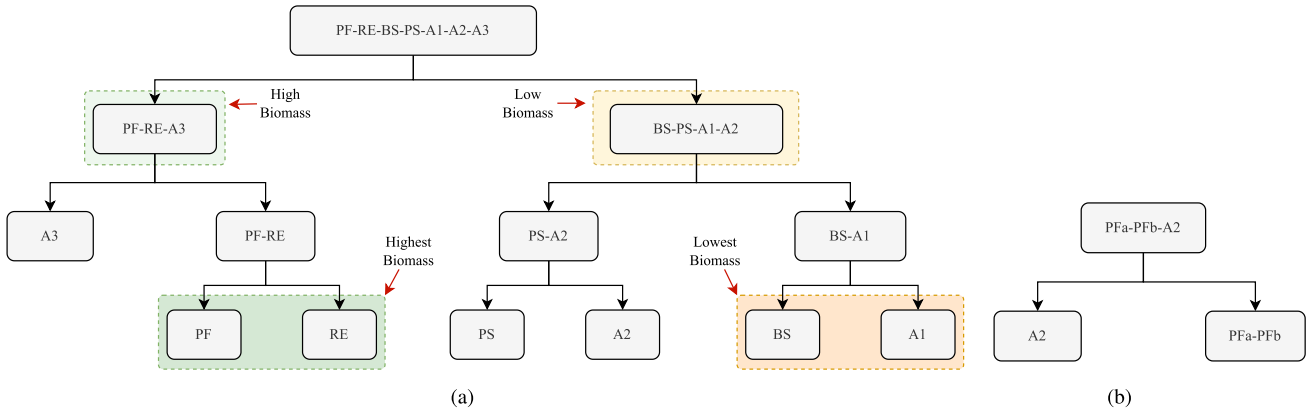


Fig. 4. Hierarchical structure derived from Tables III and VI. (a) Case I. (b) Case II.

TABLE III

SEPARABILITY ANALYSIS RESULTS FOR CASE I WITH $\rho = 5$ AND $\gamma = 1.0$. ONLY THE SMALLEST p-VALUES ARE REPORTED

Groups/Bi-partitions	W	T	W/T	p-value
(A3,PF,RE)-(A1,PS,BS,A2)	7067.3	29683.4	0.238	0.0005
(A3,PF,PS)-(A1,RE,BS,A2)	7002.8	28868.4	0.243	0.0005
(A1,BS,A2)-(A3,PF,PS,RE)	7732.3	30473.9	0.254	0.0005
(A1,PS,BS)-(A3,PF,RE,A2)	7880.2	29020.9	0.272	0.0005
(PF,PS,RE)-(A1,A3,BS,A2)	7968.5	29136.5	0.273	0.0005
(A1,RE,BS)-(A3,PF,PS,A2)	8310.6	28849.9	0.288	0.0005
(A3,PF)-(A1,PS,RE,BS,A2)	9241.3	27288.1	0.339	0.0005
(A3,BS)-(A1,PF,PS,RE,A2)	9687.6	24046.9	0.403	0.0005
(A3,A2)-(A1,PF,PS,RE,BS)	10489.6	24990.6	0.420	0.0005
(A3,PS)-(A1,PF,RE,BS,A2)	10345.3	24297.5	0.426	0.0005
(A1,A3)-(PF,PS,RE,BS,A2)	10985.4	24032.2	0.457	0.0005
(A3,RE)-(A1,PF,PS,BS,A2)	11104.7	24286.1	0.457	0.0005
(A1,BS)-(A3,PF,PS,RE,A2)	11915.6	24881.2	0.479	0.0005
(BS,A2)-(A1,A3,PF,PS,RE)	11698.6	24029.9	0.487	0.0005
(A3)-(A1,PF,PS,RE,BS,A2)	16536.7	20334.8	0.813	0.0005
(BS)-(A1,A3,PF,PS,RE,A2)	18742.5	16068.5	1.166	0.0005
(PF)-(A1,A3,PS,RE,BS,A2)	18959.8	15561.6	1.218	0.0005
(A3)-(PF,RE)	1346.0	6617.8	0.203	0.0005
(PF)-(A3,RE)	1835.6	5385.0	0.341	0.0005
(RE)-(A3,PF)	2245.1	4586.4	0.490	0.0005
(PS,A2)-(A1,BS)	1811.4	6918.0	0.262	0.0005
(BS)-(A1,PS,A2)	3077.7	5902.0	0.521	0.0005
(PS)-(A1,BS,A2)	3127.9	5818.1	0.538	0.0005
(RE)-(PF)	391.7	1581.8	0.202	0.0005
(BS)-(A1)	400.4	1504.6	0.2666	0.0005
(A2)-(PS)	449.6	1593.0	0.282	0.001

TABLE IV

p-VALUES OBSERVED IN DIFFERENT COMBINATIONS OF ρ AND γ

	ρ							
γ	1	2	3	4	5	6	7	8
0.1	0.53	0.47	0.51	0.48	0.48	0.48	0.49	0.46
0.2	0.56	0.58	0.55	0.46	0.48	0.45	0.47	0.46
0.3	0.44	0.62	0.53	0.47	0.45	0.52	0.45	0.45
0.4	0.16	0.52	0.57	0.60	0.57	0.52	0.43	0.45
0.5	0.49	0.61	0.58	0.40	0.47	0.45	0.36	0.44
0.6	0.67	0.49	0.45	0.42	0.41	0.47	0.38	0.40
0.7	0.43	0.14	0.66	0.67	0.34	0.45	0.39	0.41
0.8	0.14	0.39	0.46	0.46	0.45	0.54	0.37	0.37
0.9	0.83	0.47	0.54	0.33	0.50	0.56	0.44	0.38
1.0	0.66	0.63	0.64	0.55	0.38	0.55	0.39	0.34
1.1	0.34	0.15	0.20	0.43	0.37	0.69	0.30	0.34
1.2	0.54	0.12	0.45	0.70	0.32	0.50	0.37	0.33
1.3	0.58	0.61	0.32	0.45	0.44	0.51	0.55	0.31
1.4	0.91	0.48	0.39	0.71	0.39	0.76	0.39	0.27
1.5	0.87	0.22	0.82	0.48	0.71	0.80	0.40	0.30
1.6	0.67	0.14	0.36	0.39	0.54	0.43	0.36	0.24
1.7	0.53	0.58	0.55	0.49	0.32	0.59	0.41	0.26
1.8	0.11	0.40	0.86	0.46	0.36	0.56	0.30	0.23

with their counterparts A1, A2, PS, and BS. In a subsequent recursive level of analysis, the group composed of A3, PF, and RE is found to be nonhomogeneous, as indicated by a p-value below the adopted significance level. The computed statistics identify the bipartition (A3)-(PF, RE), corresponding to $(\boldsymbol{X}_L)$–$(\boldsymbol{X}_R)$, as the configuration that minimizes within-group variability while maximizing between-group variability. Similarly, the group formed by A1, A2, PS, and BS is biparted into (PS, A2)-(A1, BS). After applying the same process at one additional recursive level, the final separation is achieved, where each class/group is fully isolated from the others. The hierarchical structure revealed by this process is depicted in Fig. 4(a).

In addition, it is worth highlighting that the structure shown in Fig. 4(a) is also consistent from an environmental perspective. At the first level, it can be observed that PF, RE, and A3 (crop at an advanced stage of growth) are not similar to the low-biomass classes A1/A2 (crops at initial/intermediate stages of growth), BS, and PS. Subsequently, in the left branch, A3 is separated from the forestry classes PF and RE. Similarly, in the right branch, the reasonable association between the intermediate crop class (A2) and PS is preserved while distinguished from the lowest biomass classes BS and A1. Finally, each class is distinctly separated from its previously associated relational pair.

Concerning Case II, due to the nonseparability between PFa and PFb, the statistic T exhibits an opposite behavior compared to Case I. This is evident in Table IV, where all p-values are greater than 5%, independent of the adopted parameter γ or ρ. When comparing the value of T shown in Table V, under a fixed $\rho = 5$, to different quantiles of the distribution resulting from the Monte Carlo simulation, it can be observed that this statistic does not display extreme behavior. Fig. 3(b) further illustrates this behavior, assuming $\rho = 5$ and $\gamma = 1.0$.

TABLE V

T AND CRITICAL VALUES OBSERVED FOR CASE II WITH $\rho = 5$

		Lower and upper quantiles						
γ	T	**0.01**	**0.025**	**0.05**	**0.50**	**0.95**	**0.975**	**0.99**
0.1	−0.02	−25.35	−21.72	−17.81	−0.82	20.48	25.77	30.17
0.2	−0.10	−22.32	−19.52	−16.82	−0.85	19.18	23.87	29.47
0.3	−0.21	−22.65	−18.92	−16.07	−0.63	18.89	24.04	29.53
0.4	−0.32	−21.00	−18.06	−15.69	−0.51	18.26	21.91	26.13
0.5	−0.44	−19.84	−17.75	−15.31	−0.32	18.49	21.99	26.07
0.6	−0.55	−20.40	−17.84	−15.72	−0.79	18.02	21.63	25.73
0.7	−0.66	−20.67	−17.57	−15.27	−0.48	17.29	20.81	25.43
0.8	−0.77	−20.61	−17.94	−15.58	−0.78	17.85	21.93	27.14
0.9	−0.88	−21.24	−18.24	−15.33	−0.72	17.31	21.79	26.77
1.0	−1.00	−20.45	−17.78	−15.17	−0.30	18.00	22.06	26.44
1.1	−1.11	−21.71	−18.16	−15.47	−0.57	17.38	21.86	26.21
1.2	−1.24	−20.64	−17.56	−15.08	−0.93	19.16	23.44	26.97
1.3	−1.37	−20.89	−18.87	−16.43	−0.75	18.37	23.70	27.98
1.4	−1.52	−21.71	−18.77	−16.06	−0.73	18.73	23.62	28.35
1.5	−1.68	−21.76	−19.01	−16.19	−0.80	20.31	25.00	29.37
1.6	−1.86	−22.18	−19.95	−17.22	−0.56	21.46	24.59	29.20
1.7	−2.05	−23.81	−20.85	−18.23	−1.14	19.91	25.44	29.70
1.8	−2.26	−24.52	−22.31	−18.91	−0.23	20.52	24.98	32.34

TABLE VI

SEPARABILITY ANALYSIS RESULTS FOR CASE II WITH $\rho = 5$ AND $\gamma = 1.0$. ONLY THE SMALLEST p-VALUES ARE REPORTED

Groups	W	T	W/T	p-value
(A2)-(PFa,PFb)	1421.3	4822.8	0.295	0.0005
(PFa)-(PFb,A2)	1676.6	4292.6	0.391	0.11
(PFb)-(PFa,A2)	1685.4	4267.8	0.395	0.13
(PFa)-(PFb)	462.034	984.555	0.469	0.77

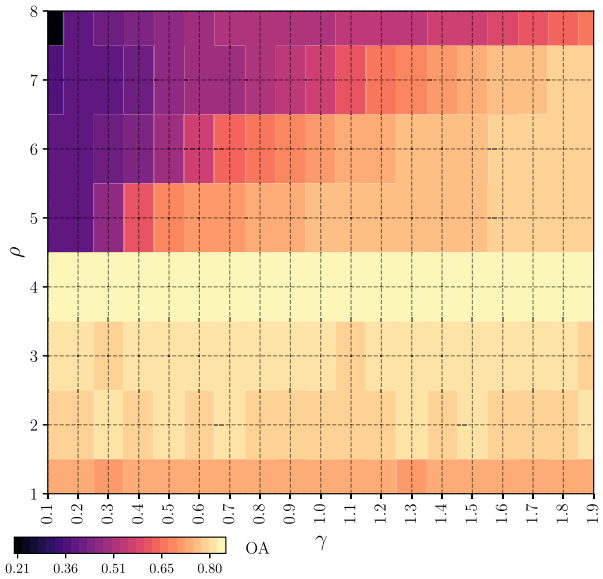
Extending the investigations of Case II by applying the hierarchical analysis of separability, with $\rho = 5$ and $\gamma = 1.0$, it is verified through Table VI that A2 is separable from PFa/b at a significance level of 5%. However, it is not possible to segregate PFa from PFb. The hierarchical structure identified is depicted in Fig. 4(b).

D. Classification

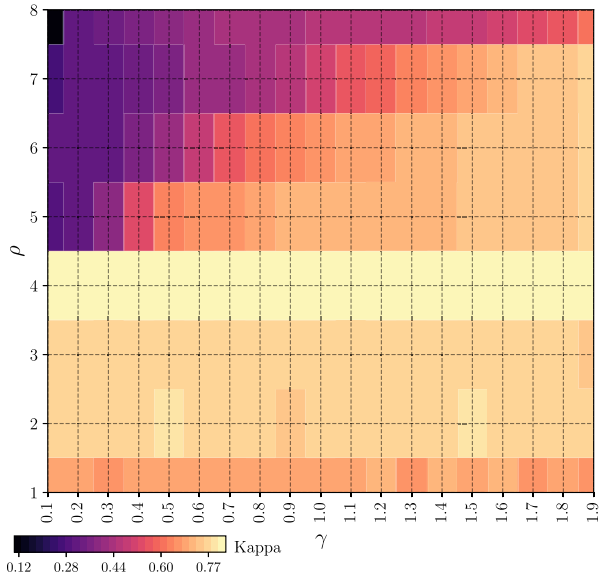
Beyond verifying the effectiveness of the proposed data analysis procedures, the results presented in Section IV-C also allow us to conclude about the compactness and separability of the different analyzed LULC classes. This set of samples is now employed as a training set for the U-statistic-based image classifier, proposed in Section III-B, and as a test set for measuring the accuracy of the results. The solid and void polygons in Fig. 2(c) are adopted for training and testing purposes, respectively.

Fig. 5 presents an ablation analysis of the impacts of γ and ρ on the classification accuracy, measured in terms of OA and kappa coefficient. According to both metrics, it is evident that the parameter ρ has a greater influence than γ. The most accurate results are achieved with $\rho = 4$ (corresponding to a 9×9 pixel neighborhood) and $\gamma = 1.5$, yielding OA and kappa coefficient values of 0.842 and 0.811, respectively. Conversely, combinations of small γ values with large ρ values result in lower performances.

Regarding the methods adopted as baselines, the MSDC and MLP classifiers achieve the most competitive results, with



(a)



(b)

Fig. 5. Classification performance in terms of (a) OA and (b) kappa coefficient for distinct ρ and γ values.

TABLE VII

CLASSIFICATION ACCURACY SUMMARY

Method	**OA**	**Kappa**	**Variance of Kappa**
MLC	0.584	0.506	1.283×10^{-5}
MDC	0.557	0.478	1.281×10^{-5}
MSDC	0.766	0.718	0.954×10^{-5}
SVM	0.589	0.502	1.237×10^{-5}
MLP	0.603	0.521	1.241×10^{-5}
Proposed	0.842	0.811	0.712×10^{-5}

OA values of 0.766 and 0.603 and kappa coefficients of 0.718 and 0.521, respectively. Table VII summarizes the classification accuracies obtained by each analyzed method. Beyond

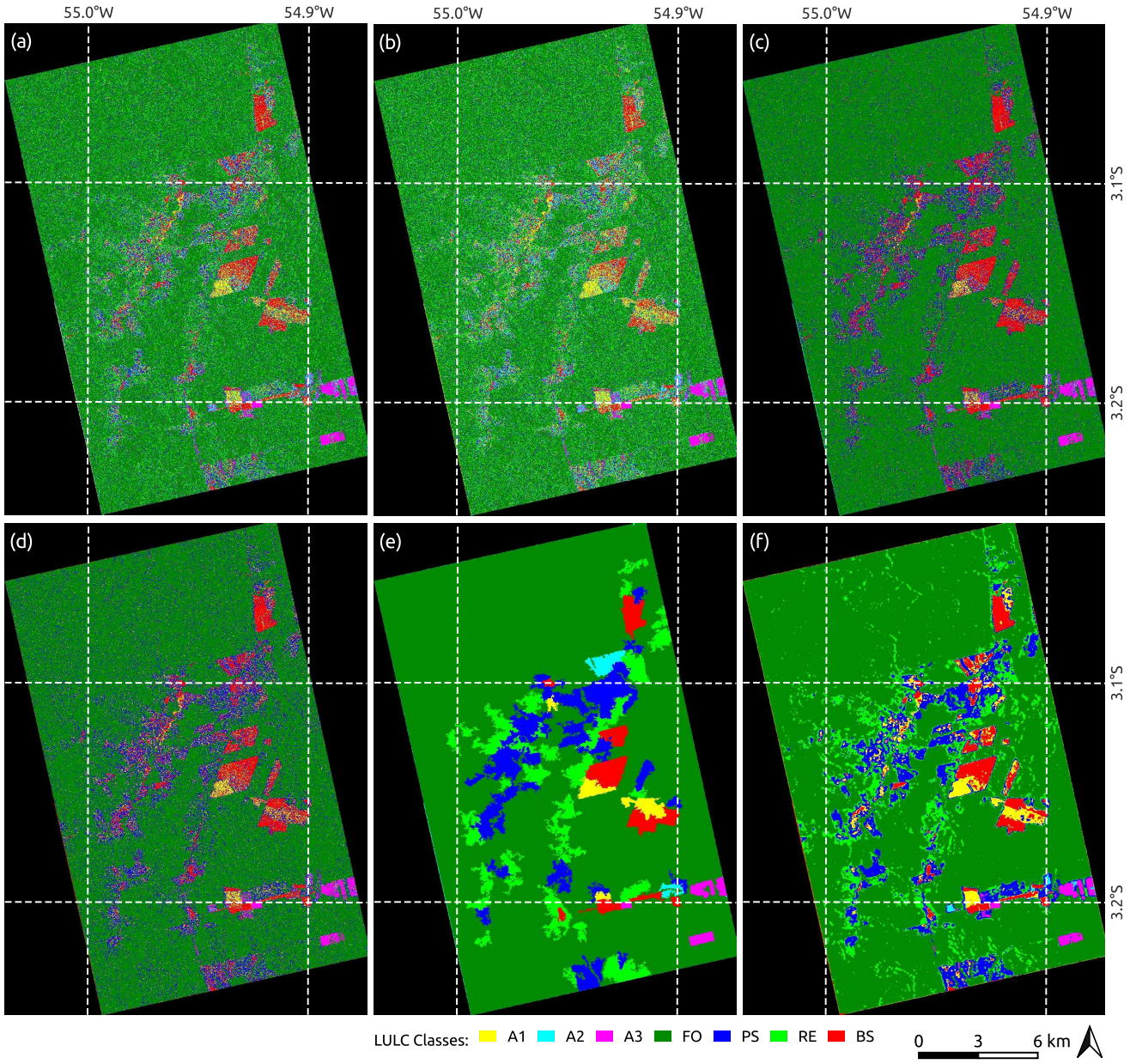


Fig. 6. Classification maps obtained by (a) MLC, (b) MDC, (c) MSDC, (d) SVM, (e) MLP, and (f) proposed methods.

its numerical superiority, the proposed approach is further validated through a significance test based on the kappa coefficient and its variance [40]. In all comparisons between the proposed method and its competitors, the observed p-values are consistently below 10^{-4}, underscoring its effectiveness.

Beyond the quantitative assessment, the classification results obtained by the analyzed methods are presented in Fig. 6. The proposed method demonstrates spatially consistent results in identifying distinct LULC types while producing more regularized (i.e., less noisy) classification maps. This outcome highlights the effectiveness of combining U-statistics, a robust tool for data analysis, with the structured treatment of observations as neighborhoods. In contrast, pixel-based approaches, such as MLC, MDA, SVM, and MLP exhibit higher confusion among certain LULC types (e.g., among PF, RE, and PS, as well as between A1 and BS), resulting in less regularized outputs. The region-based MSDC approach provides a comparatively improved mapping by incorporating spatial variability through a prior segmentation process, enhancing its ability to effectively delineate LULC classes.

V. Conclusion

This study introduced the application of nonparametric U-statistics for hypothesis testing, separability analysis, and classification of PolSAR data by considering the covariance matrix intensities. Although statistical models exist for these intensities, they are convoluted and prone to numerical instabilities.

The proposed procedures were validated with experiments using actual ALOS-PALSAR imagery from the Amazonian region. The results demonstrated that the proposed methods

are powerful tools for analyzing observations from PolSAR data.

The approach based on U-statistics is general and can be used in other situations regardless of the underlying model.

As directions for future research, we propose the following: 1) the development of strategies for the automatic estimation of the parameters σ and ρ; 2) the extension of the analysis to additional datasets, including those acquired by multispectral, hyperspectral, and LiDAR sensors; and 3) the exploration of integrating U-statistics within deep learning frameworks to enhance the PolSAR data analysis.

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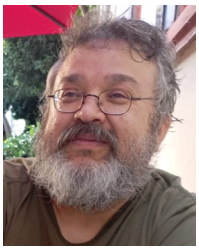
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